Blancmange Analysis (Stronger Version)

Dyadic Fold Geometry, Faber–Schauder Approximation, and Fold-Regular Learning

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Abstract

This paper develops *Blancmange Analysis*, a dyadic alternative to the usual locally linear viewpoint of classical real analysis. Instead of taking Euclidean increments as primitive, the framework studies functions through recursive dyadic folds generated by the tent map

$$\psi(x) = \operatorname{dist}(x, \mathbb{Z}),$$

and by the Takagi–Landsberg family

$$B_\lambda(x) = \sum_{n=0}^{\infty} \lambda^n \psi(2^n x), \qquad 0 < \lambda < 1.$$

The central thesis is that the real line may be analyzed through symbolic fold addresses, fold metrics, fold-relative regularity, and localized tent expansions. A key correction is made to the naive shifted-tent system: since ψ is periodic, the expressions $\psi(2^n x - k)$ do not produce localized atoms for integer k. The correct localized objects are dyadic hat functions, equivalent to the Faber–Schauder system. We prove uniform convergence of the Takagi–Landsberg functions, reconstruction from dyadic addresses, metric validity and topological equivalence of fold metrics, density of localized tent expansions in $C[0, 1]$, and positive semidefiniteness of fold kernels. We then formulate algorithms for numerical evaluation, fold-address reconstruction, tent regression, kernel learning, and fold-regularized neural learning. The result is not a replacement for classical analysis, but a rigorous dyadic fold calculus for rough functions, multiscale approximation, statistics, and machine learning.

The paper ends with "The End"

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1 Introduction

Classical analysis studies functions by comparing their local variation with straight Euclidean displacement. The canonical infinitesimal object is the quotient

$$\frac{f(x+h) - f(x)}{h},$$

and differentiability means that this quotient stabilizes as $h \to 0$. This gives a powerful local theory whenever the underlying geometry is well approximated by straight line segments.

The blancmange function, also known as the Takagi function, challenges this intuition. It is continuous everywhere but nowhere differentiable in the classical case. Yet it is not chaotic in an arbitrary sense. It is constructed from an extremely simple recursive mechanism: repeated dyadic folding.

Let

$$\psi(x) = \mathrm{dist}(x, \mathbb{Z}).$$

The classical blancmange function is

$$B(x) = \sum_{n=0}^{\infty} 2^{-n} \psi(2^n x).$$

Each term $\psi(2^n x)$ folds the line into 2^n triangular oscillations. Thus the function is not merely pathological; it encodes a complete multiscale dyadic geometry.

The guiding principle of this paper is:

Do not analyze functions only by straightening curves; analyze them also by folding the line.

Blancmange Analysis is the study of functions, regularity, approximation, statistics, and learning relative to this dyadic fold geometry.

2 The Tent Map and the Takagi–Landsberg Family

Definition 2.1 (Tent map)**.** *The tent map is*

$$\psi(x) = \operatorname{dist}(x, \mathbb{Z}).$$

Equivalently, for $x \in [0, 1]$,

$$\psi(x) = \begin{cases} x, & 0 \le x \le \frac{1}{2}, \\ 1 - x, & \frac{1}{2} \le x \le 1, \end{cases}$$

extended periodically with period 1.

Definition 2.2 (Takagi–Landsberg family)**.** *For $0 < \lambda < 1$, define*

$$B_\lambda(x) = \sum_{n=0}^{\infty} \lambda^n \psi(2^n x).$$

The classical blancmange function is

$$B(x) = B_{1/2}(x).$$

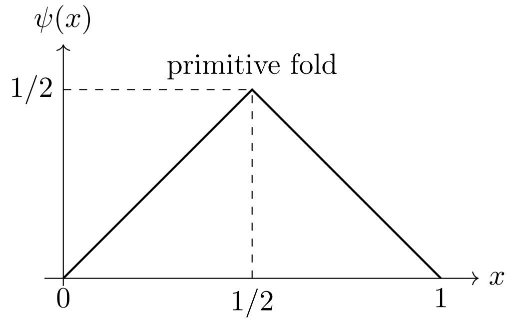

Figure 1: The primitive dyadic fold: the tent map $\psi(x) = \operatorname{dist}(x, \mathbb{Z})$ on $[0, 1]$.

Theorem 2.3 (Uniform convergence)**.** *For every $0 < \lambda < 1$, the series*

$$B_\lambda(x) = \sum_{n=0}^{\infty} \lambda^n \psi(2^n x)$$

converges uniformly on $\mathbb{R}$. Consequently B_λ is continuous.

Proof. For all $x \in \mathbb{R}$,

$$0 \le \psi(x) \le \frac{1}{2}.$$

Hence

$$|\lambda^n \psi(2^n x)| \le \frac{1}{2} \lambda^n.$$

Since

$$\sum_{n=0}^{\infty} \frac{1}{2} \lambda^n = \frac{1}{2(1 - \lambda)} < \infty,$$

the Weierstrass M-test gives uniform convergence. Each summand is continuous, so the uniform limit is continuous. □

Proposition 2.4 (Elementary regularity regimes). *If $0 < \lambda < 1/2$, then B_λ is Lipschitz. If $1/2 < \lambda < 1$, then B_λ is Hölder continuous of exponent*

$$\alpha = -\log_2 \lambda.$$

For $\lambda = 1/2$, B has the borderline modulus $O(h \log(1/h))$.

Proof. The function ψ is 1-Lipschitz. Therefore the n-th summand has Lipschitz constant $(2\lambda)^n$. If $2\lambda < 1$, the sum of these Lipschitz constants is finite, so B_λ is Lipschitz.

For $1/2 < \lambda < 1$, let $h = |x - y|$ and choose N such that

$$2^{-(N+1)} < h \le 2^{-N}.$$

Split the difference into low and high frequency parts:

$$|B_\lambda(x) - B_\lambda(y)| \le \sum_{n=0}^{N} \lambda^n |\psi(2^n x) - \psi(2^n y)| + \sum_{n=N+1}^{\infty} \lambda^n.$$

The first part is bounded by

$$h \sum_{n=0}^{N} (2\lambda)^n \le Ch(2\lambda)^N.$$

Since $\lambda = 2^{-\alpha}$, this is $O(h^\alpha)$. The tail is

$$\sum_{n=N+1}^{\infty} \lambda^n = O(\lambda^N) = O(h^\alpha).$$

The case $\lambda = 1/2$ gives the logarithmic borderline estimate. □

3 Real Numbers as Dyadic Fold Addresses

Classical analysis begins with points. Blancmange Analysis begins with addresses.

Let

$$\Sigma = \{0, 1\}^{\mathbb{N}}$$

be the space of infinite binary sequences.

Definition 3.1 (Dyadic address map). *Define*

$$\pi : \Sigma \to [0, 1], \qquad \pi(\varepsilon_1, \varepsilon_2, \ldots) = \sum_{n=1}^{\infty} \frac{\varepsilon_n}{2^n}.$$

The sequence $(\varepsilon_n)_{n \ge 1}$ is called a dyadic fold address.

For non-dyadic x, the address is unique. Dyadic rationals have two binary addresses:

$$0.1000\ldots_2 = 0.0111\ldots_2\,.$$

Thus

$$[0, 1] \cong \Sigma / \sim,$$

where $\sim$ identifies the two binary representations of each dyadic rational.

Definition 3.2 (Fold itinerary). *For $x \in [0,1]$ not dyadic, define*

$$\varepsilon_{n+1}(x) = \begin{cases} 0, & \{2^n x\} < \frac{1}{2}, \\ 1, & \{2^n x\} > \frac{1}{2}. \end{cases}$$

This records whether x lies on the rising or falling side of the n-th fold.

Proposition 3.3 (Reconstruction from fold itinerary). *For every non-dyadic $x \in [0,1]$,*

$$x = \sum_{n=1}^{\infty} \frac{\varepsilon_n(x)}{2^n}.$$

For dyadic x, the same identity holds after quotienting the two binary representatives.

Proof. At level n, the condition $\{2^{n-1}x\} < 1/2$ places x in the left half of a dyadic interval of length $2^{-(n-1)}$, while the opposite condition places it in the right half. Hence each ε_n selects a unique dyadic child interval. The nested intervals have lengths 2^{-n}, and their intersection contains exactly x. This gives the binary expansion of x. $\square$

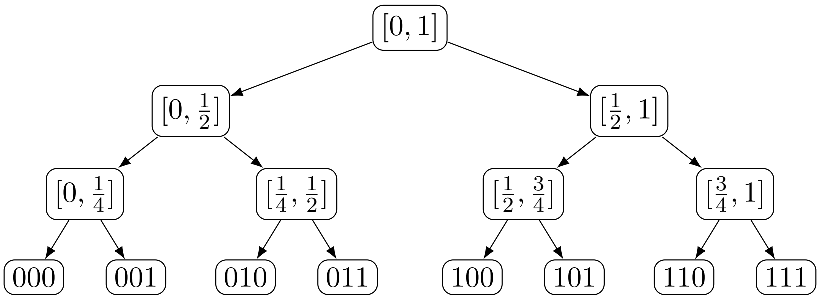

Figure 2: Dyadic fold descent: real numbers are branch addresses through nested binary folds.

4 Axioms of Blancmange Analysis

Axiom 4.1 (Fold primacy). *The primitive local probes are dyadic folds generated by the tent map ψ, rather than linear coordinate increments alone.*

Axiom 4.2 (Scale descent). *The limiting operation $h \to 0$ is replaced or supplemented by dyadic descent:*

$$2^{-n} \to 0.$$

Axiom 4.3 (Fold-relative comparison). *The variation of a function f may be measured relative to the variation of a fold observable such as B_λ, not only relative to Euclidean displacement.*

Axiom 4.4 (Localized approximation). *Approximation is performed using genuine localized dyadic hat atoms, not merely global tent waves.*

5 Fold Metrics and Fold-Regularity

Definition 5.1 (Fold metric). *For $0 < \lambda < 1$ and $\eta > 0$, define*

$$d_{\lambda,\eta}(x,y) = |x-y| + \eta |B_\lambda(x) - B_\lambda(y)|.$$

When $\lambda = 1/2$, write $d_B = d_{1/2,\eta}$.

Classical analysis	Blancmange Analysis	Interpretation
Point x	Address (ε_n)	Branch sequence
Increment $h \to 0$	Scale $2^{-n} \to 0$	Dyadic descent
Line segment	Tent segment	Folded geometry
Derivative df/dx	Fold quotient or fold slope	Comparison against folds
Smoothness	Fold regularity	Stability under dyadic folding
Taylor expansion	Tent expansion	Fold-based approximation
Fourier modes	Tent atoms	Multiscale folded features
Euclidean kernel	Fold kernel	Similarity through fold coordinates

Table 1: Dictionary between classical analysis and Blancmange Analysis.

Theorem 5.2 (Metric validity). *For every $0 < \lambda < 1$ and every $\eta > 0$, $d_{\lambda,\eta}$ is a metric on $[0,1]$.*

Proof. Non-negativity and symmetry are immediate. If $d_{\lambda,\eta}(x,y) = 0$, then $|x-y| = 0$, so $x = y$. Finally, for $x, y, z \in [0,1]$,

$$d_{\lambda,\eta}(x,z) = |x-z| + \eta|B_\lambda(x) - B_\lambda(z)|$$

$$\leq |x-y| + |y-z| + \eta|B_\lambda(x) - B_\lambda(y)| + \eta|B_\lambda(y) - B_\lambda(z)| = d_{\lambda,\eta}(x,y) + d_{\lambda,\eta}(y,z).$$

Thus $d_{\lambda,\eta}$ is a metric. □

Theorem 5.3 (Topological equivalence). *The metric $d_{\lambda,\eta}$ induces the same topology on $[0,1]$ as the Euclidean metric.*

Proof. Since

$$|x-y| \leq d_{\lambda,\eta}(x,y),$$

convergence in $d_{\lambda,\eta}$ implies Euclidean convergence.

Conversely, if $x_m \to x$ in the Euclidean metric, then continuity of B_λ gives

$$B_\lambda(x_m) \to B_\lambda(x).$$

Therefore

$$d_{\lambda,\eta}(x_m, x) = |x_m - x| + \eta|B_\lambda(x_m) - B_\lambda(x)| \to 0.$$

Thus the two metrics induce the same topology. □

Definition 5.4 (Blancmange-Hölder regularity). *A function $f : [0,1] \to \mathbb{R}$ is B_λ-Hölder of order $\alpha \in (0,1]$ if there exists $C > 0$ such that*

$$|f(x) - f(y)| \leq C d_{\lambda,\eta}(x,y)^\alpha$$

for all $x, y \in [0,1]$.

Proposition 5.5 (Classical Hölder implies fold-Hölder). *If f is ordinary Hölder of order α, then f is B_λ-Hölder of order α.*

Proof. If f is ordinary Hölder, then

$$|f(x) - f(y)| \leq C|x-y|^\alpha.$$

Since $|x-y| \leq d_{\lambda,\eta}(x,y)$, we obtain

$$|f(x) - f(y)| \leq C d_{\lambda,\eta}(x,y)^\alpha.$$

□

6 Fold Derivatives and Fold Slopes

The classical derivative compares f-increments to x-increments. A fold derivative compares f-increments to changes in a fold observable.

Definition 6.1 (Dyadic fold quotient). *Let $f : [0, 1] \to \mathbb{R}$. At an admissible point x, define the n-th dyadic fold quotient by*

$$Q_n^{B_\lambda} f(x) = \frac{f(x + 2^{-n}) - f(x)}{B_\lambda(x + 2^{-n}) - B_\lambda(x)},$$

whenever $x + 2^{-n} \in [0, 1]$ and the denominator is nonzero.

Definition 6.2 (Dyadic fold derivative). *If the limit exists along all sufficiently large admissible n, define*

$$D_{B_\lambda} f(x) = \lim_{n\to\infty} Q_n^{B_\lambda} f(x).$$

Definition 6.3 (Fold upper slope). *The fold upper slope of f at x is*

$$|\nabla_B f|(x) = \limsup_{r\downarrow 0} \sup_{0<d_{\lambda,\eta}(x,y)<r} \frac{|f(x) - f(y)|}{d_{\lambda,\eta}(x, y)}.$$

Proposition 6.4 (Chain rule for fold-observable functions). *Let $f(x) = F(B_\lambda(x))$, where F is differentiable at $B_\lambda(x)$. If $D_{B_\lambda} f(x)$ exists, then*

$$D_{B_\lambda} f(x) = F'(B_\lambda(x)).$$

Proof. For admissible n,

$$Q_n^{B_\lambda} f(x) = \frac{F(B_\lambda(x + 2^{-n})) - F(B_\lambda(x))}{B_\lambda(x + 2^{-n}) - B_\lambda(x)}.$$

Since B_λ is continuous,

$$B_\lambda(x + 2^{-n}) \to B_\lambda(x).$$

Thus the quotient converges to the derivative of F at $B_\lambda(x)$. □

7 Correct Localized Tent Atoms

A crucial correction is necessary. Since ψ has period 1,

$$\psi(2^n x - k) = \psi(2^n x)$$

for every integer k. Therefore these expressions do not produce localized atoms.

The correct localized atoms are dyadic hat functions.

Definition 7.1 (Localized dyadic tent atom). *For $n \geq 0$ and $0 \leq k < 2^n$, define*

$$\phi_{n,k}(x) = \max\left\{0,\, 1 - \left|2^{n+1}x - (2k + 1)\right|\right\}.$$

Equivalently, $\phi_{n,k}$ is the triangular hat function supported on

$$\left[\frac{k}{2^n}, \frac{k+1}{2^n}\right],$$

with peak value 1 at

$$\frac{2k+1}{2^{n+1}}.$$

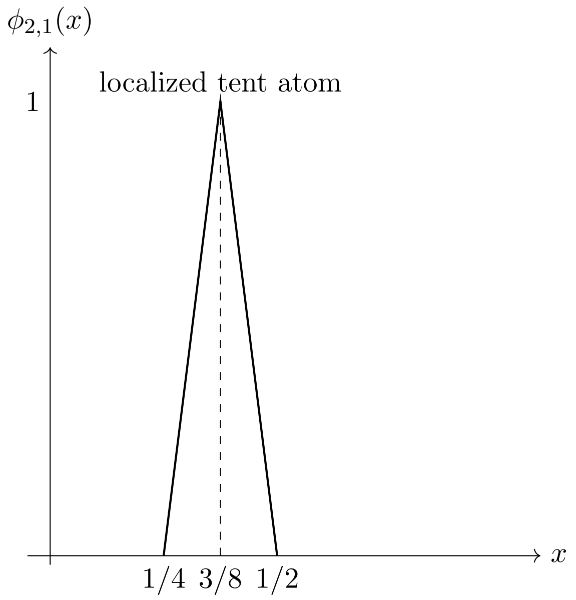

Figure 3: A localized dyadic tent atom. Unlike $\psi(2^n x - k)$, the atom $\phi_{n,k}$ is genuinely supported on one dyadic interval.

Definition 7.2 (Tent expansion)**.** *A localized tent expansion of f is an expression of the form*

$$f(x) \sim a_0 + a_1 x + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} c_{n,k}\phi_{n,k}(x).$$

Theorem 7.3 (Density of localized tent expansions)**.** *Let*

$$\mathcal{T} = \operatorname{span}\{1, x, \phi_{n,k} : n \geq 0,\ 0 \leq k < 2^n\}.$$

Then $\mathcal{T}$ is uniformly dense in $C[0,1]$.

Proof. Let $f \in C[0,1]$ and $\varepsilon > 0$. Since f is uniformly continuous on $[0,1]$, there exists N such that if $|x - y| \leq 2^{-N}$, then

$$|f(x) - f(y)| < \varepsilon.$$

Let $I_N f$ be the continuous piecewise-linear interpolation of f on the dyadic grid

$$0, \frac{1}{2^N}, \frac{2}{2^N}, \ldots, 1.$$

Then

$$\|f - I_N f\|_\infty < \varepsilon.$$

Every continuous piecewise-linear function on a dyadic grid can be expressed as a finite linear combination of 1, x, and the dyadic hat functions $\phi_{n,k}$ with $n < N$. Hence $I_N f \in \mathcal{T}$. Therefore $\mathcal{T}$ is uniformly dense in $C[0,1]$. □

8 Approximation Error and Fold Complexity

Definition 8.1 (Fold complexity)**.** *For a truncated expansion*

$$f_N(x) = a_0 + a_1 x + \sum_{n=0}^{N} \sum_{k=0}^{2^n-1} c_{n,k}\phi_{n,k}(x),$$

define the weighted fold complexity

$$\mathcal{C}_s(f_N) = \sum_{n=0}^{N} 2^{2sn} \sum_{k=0}^{2^n-1} c_{n,k}^2,$$

where $s \geq 0$ is a smoothness index.

The weight 2^{2sn} penalizes high-scale coefficients. Large s enforces stronger dyadic smoothness, while small s permits rougher fold-scale behavior.

Proposition 8.2 (Coefficient decay and regularity intuition)**.** *If the tent coefficients satisfy*

$$|c_{n,k}| \leq C 2^{-n(\alpha+1/2)}$$

uniformly in k, then the high-frequency contribution of the expansion is controlled at dyadic scale 2^{-n} with effective Hölder exponent α.

Proof. At level n, at most a bounded number of tent atoms are nonzero at any point x. Therefore the pointwise contribution of level n is bounded by a constant multiple of $2^{-n(\alpha+1/2)}$. Summing over $n \geq N$ yields a tail controlled by a geometric series. The resulting scale behavior is governed by the exponent α. □

9 Algorithms

This section gives algorithms in plain mathematical pseudocode, avoiding nonstandard algorithm packages.

Algorithm 1: Numerical Evaluation of B_λ

Input: $x \in [0,1]$, $0 < \lambda < 1$, truncation level N.
Steps:

1. Set $S \leftarrow 0$.
2. For $n = 0, 1, \ldots, N$:
   (a) Compute $u \leftarrow 2^n x - \lfloor 2^n x \rfloor$.
   (b) Compute $t \leftarrow \min\{u, 1-u\}$.
   (c) Update $S \leftarrow S + \lambda^n t$.
3. Return S.

Error bound:

$$|B_\lambda(x) - S| \leq \sum_{n=N+1}^{\infty} \frac{1}{2}\lambda^n = \frac{\lambda^{N+1}}{2(1-\lambda)}.$$

Algorithm 2: Fold-Address Reconstruction

Input: Bits $(\varepsilon_1, \ldots, \varepsilon_N)$.
Steps:

$$x_N = \sum_{n=1}^{N} \frac{\varepsilon_n}{2^n}.$$

Error bound:

$$|x - x_N| \leq 2^{-N}.$$

Algorithm 3: Least-Squares Tent Regression

Input: Data $(x_i, y_i)_{i=1}^m$, maximum scale N.
Model:

$$\widehat{f}_N(x) = \beta_0 + \beta_1 x + \sum_{n=0}^{N} \sum_{k=0}^{2^n-1} \beta_{n,k}\phi_{n,k}(x).$$

Optimization problem:

$$\widehat{\beta} = \arg\min_{\beta} \sum_{i=1}^{m} \left(y_i - \beta_0 - \beta_1 x_i - \sum_{n=0}^{N} \sum_{k=0}^{2^n-1} \beta_{n,k}\phi_{n,k}(x_i) \right)^2 .$$

Algorithm 4: Ridge-Regularized Fold Regression

Optimization problem:

$$\widehat{\beta} = \arg\min_{\beta} \left[\sum_{i=1}^{m} \left(y_i - \widehat{f}_N(x_i) \right)^2 + \rho \sum_{n=0}^{N} w_n \sum_{k=0}^{2^n-1} \beta_{n,k}^2 \right],$$

where $\rho > 0$ and w_n increases with n, for example $w_n = 2^{2sn}$.

10 Statistics and Data Science

Suppose data are generated by

$$Y_i = f(X_i) + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid X_i] = 0, \qquad \mathrm{Var}(\varepsilon_i \mid X_i) = \sigma^2.$$

Define the fold feature vector

$$\Phi_N(x) = \big(1,\, x,\, B_\lambda(x),\, \phi_{0,0}(x),\, \phi_{1,0}(x),\, \phi_{1,1}(x),\, \ldots,\, \phi_{N,2^N-1}(x)\big) .$$

The dimension is

$$p_N = 3 + \sum_{n=0}^{N} 2^n = 2^{N+1} + 2.$$

A linear fold model is

$$\widehat{f}_N(x) = \theta^\top \Phi_N(x).$$

Theorem 10.1 (Sieve consistency principle)**.** *Assume $f \in C[0,1]$, $\mathbb{E}[\varepsilon_i \mid X_i] = 0$, and $\mathbb{E}[\varepsilon_i^2] < \infty$. Let $N = N_m \to \infty$ such that*

$$\frac{p_{N_m}}{m} \to 0.$$

Under standard full-rank design assumptions, the least-squares fold estimator is consistent in mean square:

$$\mathbb{E}\left[(\widehat{f}_{N_m}(X) - f(X))^2\right] \to 0.$$

Proof. The risk decomposes into approximation error and estimation error:

$$\mathbb{E}[(\widehat{f}_N(X) - f(X))^2] \le 2\mathbb{E}[(\widehat{f}_N(X) - f_N^*(X))^2] + 2\mathbb{E}[(f_N^*(X) - f(X))^2],$$

where f_N^* is the best L^2 approximation to f in the span of Φ_N. By density of localized tent expansions, the approximation error tends to zero as $N \to \infty$. Under the condition $p_N/m \to 0$ and standard full-rank assumptions, the estimation error tends to zero. Hence the total risk tends to zero. □

Task	Classical tool	Blancmange tool
Regression	Polynomials, splines	Localized tent regression
Classification	Kernel machines	Fold kernels
Denoising	Wavelets, filters	Fold-scale shrinkage
Feature engineering	Fourier features	Fold-address features
Anomaly detection	Residual thresholds	Fold-scale residuals
Time series	ARMA, spectra	Dyadic fold decomposition
Representation learning	Neural embeddings	Fold-augmented embeddings

Table 2: Statistical and machine-learning uses of blancmange features.

11 Fold Kernels

Definition 11.1 (Finite fold kernel). *For nonnegative weights γ_j, define*

$$K_N(x,y) = \sum_{j=1}^{p_N} \gamma_j \Phi_{N,j}(x)\Phi_{N,j}(y).$$

Theorem 11.2 (Positive semidefiniteness). *If $\gamma_j \geq 0$, then K_N is positive semidefinite.*

Proof. For any $x_1, \ldots, x_m \in [0,1]$ and $c_1, \ldots, c_m \in \mathbb{R}$,

$$\sum_{i=1}^{m}\sum_{\ell=1}^{m} c_i c_\ell K_N(x_i, x_\ell) = \sum_{j=1}^{p_N} \gamma_j \left(\sum_{i=1}^{m} c_i \Phi_{N,j}(x_i)\right)^2 \geq 0.$$

Therefore K_N is positive semidefinite. □

Definition 11.3 (Infinite localized fold kernel). *Let $\gamma_{n,k} \geq 0$. Define*

$$K(x,y) = \gamma_{-1} + \gamma_0 xy + \sum_{n=0}^{\infty}\sum_{k=0}^{2^n-1} \gamma_{n,k}\phi_{n,k}(x)\phi_{n,k}(y),$$

whenever the series converges.

Proposition 11.4 (Infinite kernel positivity). *If*

$$\sum_{n=0}^{\infty}\sum_{k=0}^{2^n-1} \gamma_{n,k} < \infty,$$

then K is a continuous positive semidefinite kernel.

Proof. Since $0 \leq \phi_{n,k}(x) \leq 1$, absolute convergence follows from the summability condition. Positive semidefiniteness follows by applying the finite-kernel argument to partial sums and passing to the limit. □

12 Fold-Regular Machine Learning

A fold-aware neural model augments the input with dyadic fold features:

$$z_N(x) = \left(x,\, B_\lambda(x),\, \phi_{0,0}(x), \ldots, \phi_{N,2^N-1}(x)\right).$$

A feedforward predictor is

$$\widehat{y} = F_\theta(z_N(x)).$$

To encourage fold-regularity, define the empirical fold-regularized loss

$$\mathcal{L}(\theta) = \frac{1}{m}\sum_{i=1}^{m} \ell(F_\theta(z_N(x_i)), y_i) + \rho \sum_{i,j=1}^{m} \frac{|F_\theta(z_N(x_i)) - F_\theta(z_N(x_j))|^2}{d_{\lambda,\eta}(x_i,x_j)^2 + \delta},$$

where $\rho > 0$ and $\delta > 0$.

This regularizer penalizes large output variation between points that are close in fold geometry.

Component	Classical ML analogue	Fold-regular analogue
Input	Raw coordinate x	Fold feature $z_N(x)$
Basis	Polynomial/spline basis	Localized dyadic tent basis
Kernel	RBF or polynomial kernel	Fold kernel
Regularization	Euclidean smoothness	Fold-metric smoothness
Neural layer	Dense feature map	Fold-augmented feature map
Graph penalty	Euclidean graph Laplacian	Fold-distance graph Laplacian

Table 3: Machine-learning translation table for fold-regular learning.

13 A Fold Graph View

Given data points $x_1, \dots, x_m$, define a fold graph with weights

$$W_{ij} = \exp\left(-\frac{d_{\lambda,\eta}(x_i,x_j)^2}{\tau^2}\right),$$

where $\tau > 0$. Let D be the diagonal degree matrix,

$$D_{ii} = \sum_{j=1}^{m} W_{ij},$$

and define the graph Laplacian

$$L = D - W.$$

For fitted values $u_i = \widehat{f}(x_i)$,

$$u^\top L u = \frac{1}{2}\sum_{i,j=1}^{m} W_{ij}(u_i - u_j)^2.$$

Thus fold-regularization can be written as a graph Laplacian penalty:

$$\min_u \sum_{i=1}^{m} (y_i - u_i)^2 + \rho u^\top L u.$$

14 Interpretation

Blancmange Analysis should not be interpreted as a rejection of real analysis. Rather, it changes the primitive geometry.

Classical real analysis begins from local straightness and treats roughness as exceptional. Blancmange Analysis begins from recursive dyadic folding and treats smoothness as a low-complexity special case.

Its conceptual components are:

$$\text{addresses} + \text{fold metrics} + \text{localized tents} + \text{fold derivatives} + \text{fold kernels} + \text{fold-regular learning}.$$

The framework is especially natural for:

1. functions with dyadic or hierarchical structure;
2. rough continuous signals;
3. multiscale regression;
4. fractal feature engineering;
5. nonparametric statistics;
6. kernel learning;
7. neural models with structured geometric inductive bias.

15 Limitations and Open Problems

Several questions remain open.

1. Develop a full Banach-space theory of B_λ-regular functions.
2. Characterize the exact relation between fold-Hölder and Besov regularity.
3. Establish minimax rates for tent regression under fold-complexity constraints.
4. Determine when fold derivatives exist for nontrivial function classes.
5. Extend the theory from dyadic folds to q-adic folds.
6. Generalize the framework to multidimensional product folds.
7. Study stochastic processes indexed by fold address space.
8. Build deep architectures whose internal layers preserve fold geometry.

16 Conclusion

This paper developed Blancmange Analysis as a rigorous dyadic fold framework for analysis, approximation, statistics, and learning. The real line was represented through binary fold addresses, the Takagi–Landsberg family supplied a canonical fold observable, and the metric

$$d_{\lambda,\eta}(x, y) = |x - y| + \eta|B_\lambda(x) - B_\lambda(y)|$$

gave a fold-enriched geometry.

A central correction was made: integer-shifted tents $\psi(2^n x - k)$ are not localized because of periodicity. The correct localized atoms are dyadic hat functions $\phi_{n,k}$, which generate the Faber–Schauder approximation system and are dense in $C[0, 1]$.

The resulting theory connects rough analysis, dyadic approximation, nonparametric regression, positive semidefinite kernels, graph regularization, and neural feature design. Blancmange Analysis is therefore best understood as a fold-relative calculus of roughness and learning.

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Glossary

Blancmange function The classical Takagi function

$$B(x) = \sum_{n=0}^{\infty} 2^{-n}\psi(2^n x).$$

Tent map The function

$$\psi(x) = \mathrm{dist}(x, \mathbb{Z}),$$

which forms triangular waves over unit intervals.

Dyadic fold The observation of x through $\psi(2^n x)$, producing 2^n triangular folds.

Fold address The binary sequence (ε_n) determining the nested dyadic location of x.

Takagi–Landsberg family The family

$$B_\lambda(x) = \sum_{n=0}^{\infty} \lambda^n \psi(2^n x), \qquad 0 < \lambda < 1.$$

Fold metric A metric of the form

$$d_{\lambda,\eta}(x, y) = |x - y| + \eta |B_\lambda(x) - B_\lambda(y)|.$$

Fold derivative A derivative-like object comparing increments of f to increments of B_λ, rather than to increments of x.

Localized tent atom A dyadic hat function $\phi_{n,k}$ supported on a dyadic interval.

Tent expansion An expansion of a function in terms of 1, x, and localized dyadic tent atoms.

Fold kernel A positive semidefinite kernel constructed from fold features.

Fold-regular learning Machine learning in which features, kernels, or regularizers are defined using dyadic fold geometry.

The End